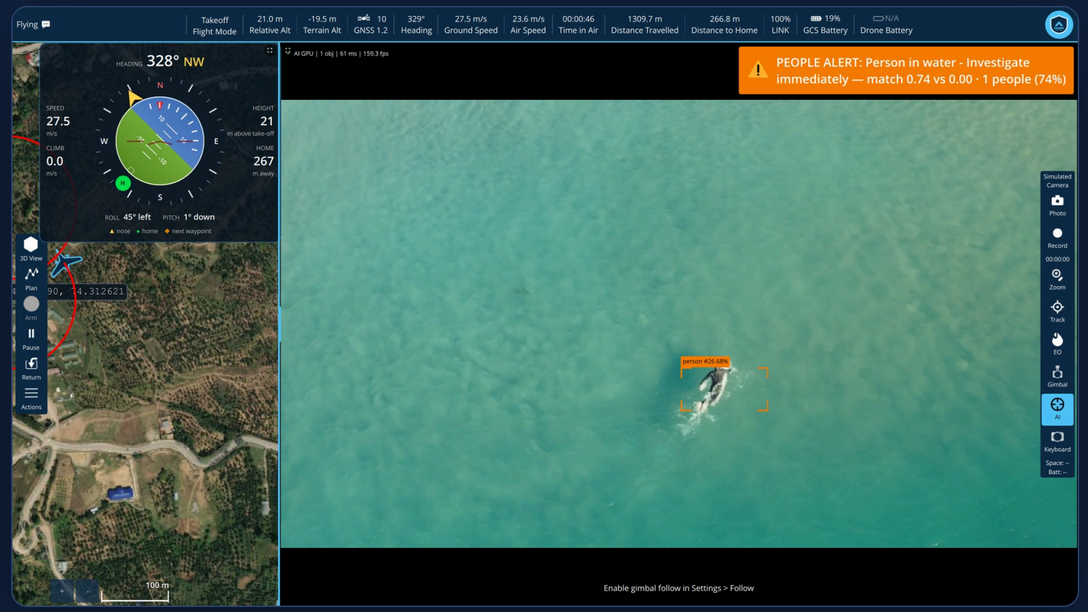
INVESTOR PITCH · OCTOBER 2026

# SKYSHIELD

## AI-based video analytics system for advanced search and rescue and ISR

Ravin tailsitter VTOL + SkyShield ground station + on-edge AI




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THE PROBLEM

# The human vigilance bottleneck

A person watching a screen for rare events misses more the longer the watch lasts. In the study that first measured it, detection accuracy fell by about 10% within the first 30 minutes of a two-hour watch.

Aircraft now stay on station for hours. The limit on long-endurance surveillance is not flight time. It is how long one operator can reliably watch.

**30 min** of watching, and detection accuracy is already down by about 10%

**11,098 km** of Indian coastline to keep under watch

**300,000** people drowned worldwide in 2021

Sources: Mackworth, Quarterly Journal of Experimental Psychology, 1948 · Government of India revised coastline length, 2024 · WHO Global status report on drowning prevention, 2024

THE SOLUTION

# One integrated system: aircraft, ground station and AI

## SkyShield GCS

**GROUND STATION · RELEASED, V2.0.2**

One interface for flight control, mission planning, live video, maps, telemetry and AI alerts.

## AI video analytics

**INTELLIGENCE LAYER · RELEASED**

Watches the live video continuously, identifies mission events, not just objects, and alerts only when it matters.

## Ravin

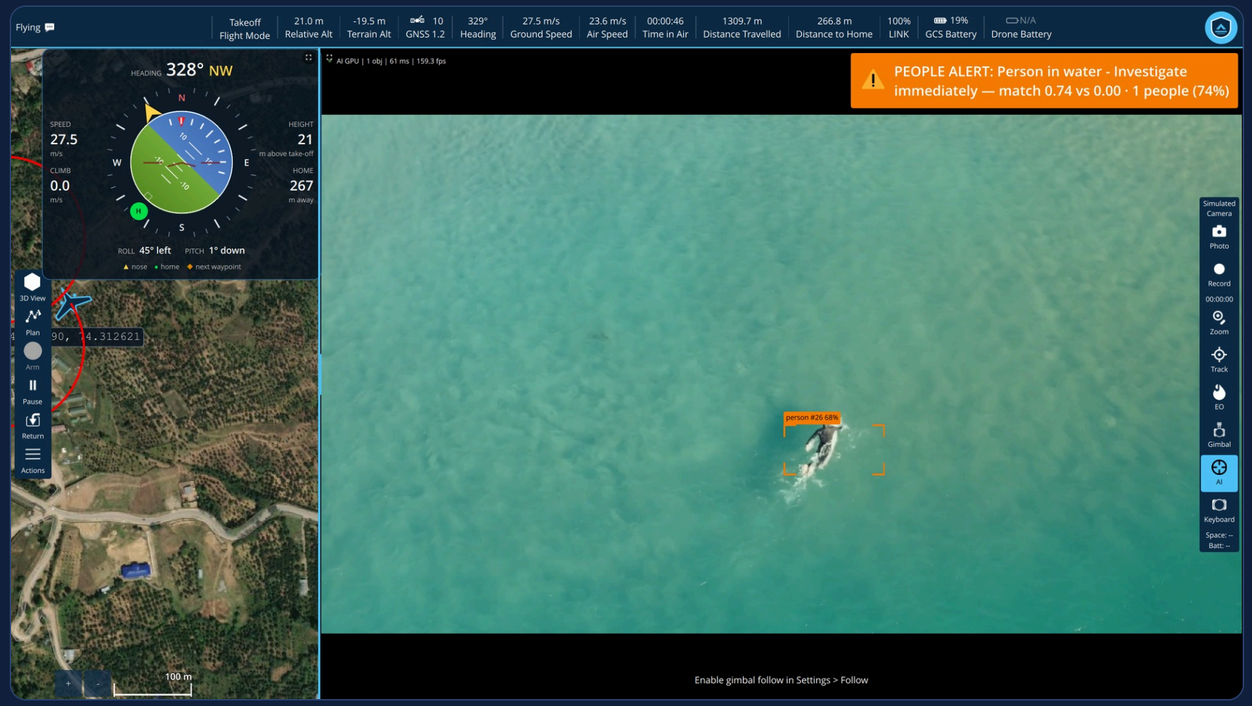
**TAILSITTER VTOL · DESIGNED, READY TO BUILD**

Takes off vertically with no runway, then cruises on its wing for range and endurance.

Engineered together as one system, not hardware with third-party software attached

THE PRODUCT, TODAY

# SkyShield GCS: built, released and demonstrable




- **Unobstructed live video** beside the map, with no floating widgets over the picture
- **Fast onboarding:** a clean interface and a built-in training simulator
- **Keyboard control mapping** for gimbal and payload, so no separate joystick hardware
- **Shipped as a product:** installer, automatic updates, device licensing, 55-page manual

Screenshot: SkyShield with a simulated aircraft and recorded aerial video, raising a Person in water alert. Not yet flown on our own aircraft.

THE INTELLIGENCE LAYER

# Purpose-built AI that reports events, not just objects

**47**

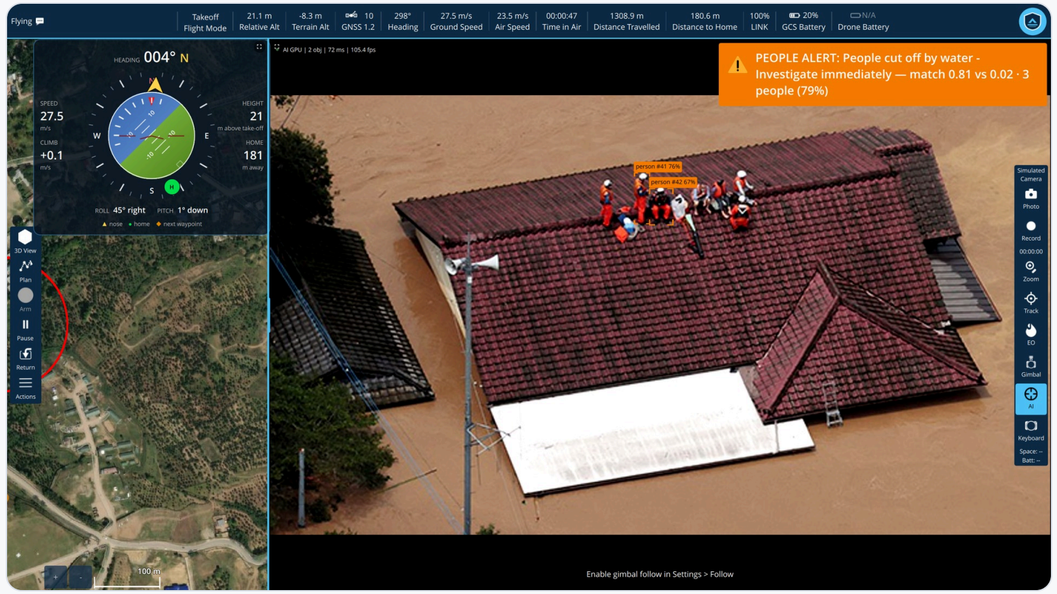
mission events in 11 categories: people, fire, flood, vehicles, routes, damage, search clues, hazards

**1**

alert at a time, spoken aloud, with the evidence and a count of people and vehicles in the area

**0**

cloud or cellular dependency: everything runs on the ground station




Alert on screen: **People cut off by water.** The people on the roof are boxed and counted.

An alert fires only when the scene matches the event clearly more than its look-alikes. Tested on recorded aerial footage so far.

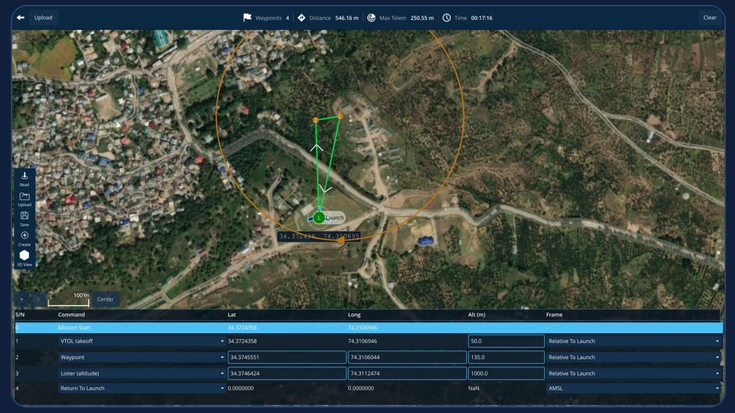
BUILT SINCE THE LAST DECK

# Three features that make it a field tool



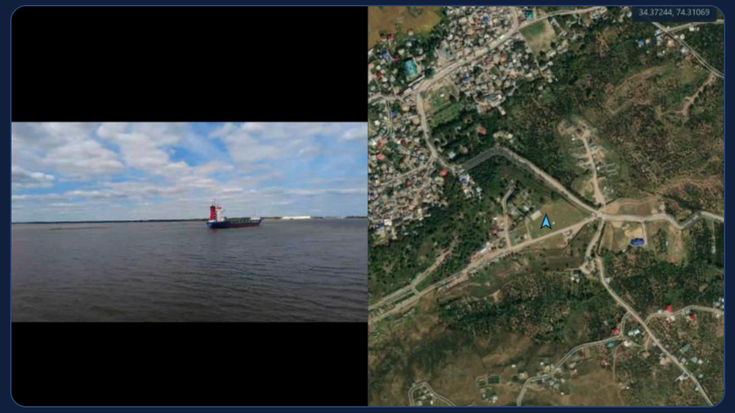

## 3D terrain view

See the real flight profile over real terrain before take-off, and follow the aircraft in 3D in flight.




## Terrain self-check

The whole planned path is checked against ground height. Anything that would hit terrain shows in red.




## Remote video terminal

Anyone on site scans a code and watches video, map and alerts in a browser. Nothing to install.

Also new: flight replay, training simulator, self-updating installs. Remote viewing over the internet needs a VPN today.

THE PLATFORM

# Ravin: endurance of a wing, footprint of a multirotor



**150 min**
endurance

**13 m/s**
wind resistance

**15 km**
operating range

| Parameter | Design value |
|---|---|
| Configuration | Tailless tailsitter VTOL, twin electric motors |
| Propulsion | Same motors for hover and cruise: no tilt, no idle lift motors |
| Endurance | 150 min with a 40 Ah battery |
| Wind resistance | Up to 13 m/s |
| Range | 15 km from the ground station |
| Fuselage | New streamlined body with a separate canopy |
| Deployment | One operator, under 3 minutes to launch |
| Manufacture | Build packages split across three vendors |

Render of design V1.30; motors and propellers not shown. Design values, to be proven in flight in Phase 1.

WHY IT WINS

# Technology and commercial advantages

### Persistent attention

The system carries the burden of watching, so hours of flight still produce awareness.

### Fully on-edge

No cloud, no cellular. It works where no network exists.

### One operator

No runway, no launch gear, minimal training. Fewer people on site.

### Lower bill of materials

A tailsitter needs no tilt servos, boom arms or separate lift motors.

### AI off the aircraft

Analytics run on the laptop, not a costly onboard computer: less weight, power and unit cost.

### No control hardware

Flight, gimbal and payload map to a standard laptop keyboard.

Manufacturing is split across independent vendors, so no single supplier holds the complete design.

WHERE IT FITS

# Four markets with the same need

## SAR

**Search and rescue**

Fast aerial search over large or difficult terrain, with alerts for people and signs of them.

## ISR

**Surveillance and reconnaissance**

Continuous watch of critical zones and remote areas, with activity reported to the operator.

## Defence

**Defence and border**

Lightweight, runway-free surveillance where connectivity is limited or denied.

## Industrial

**Mining and site safety**

The original use case: monitoring large, high-risk sites without constant manual watching.

The aircraft class is already bought commercially: tailsitter mapping VTOLs are on sale today. SkyShield adds what they lack, on-edge AI alerting built into the ground station.

Market size and order of entry follow on the next slide

TARGET MARKET

# India first: defence, then disaster response, then industry

**$0.47 bn to $1.39 bn**
India's drone market, 2025 to 2030

**16,756 a year**
drones procured in India by 2030

**$23 bn**
India's drone manufacturing potential by 2030

1 **Defence and paramilitary surveillance**
The largest segment of the market. Entry through iDEX, where the application is already submitted.

2 **Disaster response and search and rescue**
National and state response forces. Flood, fire and person-in-distress alerts are already built.

3 **Mining and industrial site safety**
The original use case. Private buyers with shorter purchase cycles than government.

Sources: MarketsandMarkets, India Drone Market (2025 to 2030); EY-FICCI, Making India the drone hub of the world (2022)

TRACTION

# What has been achieved since July

| When | Milestone |
|---|---|
| Jul 2026 | Ground station interface complete on an open-source flight-control base |
| Aug 2026 | On-device detection and tracking, 3D terrain view, device-bound licensing |
| Aug 2026 | Aircraft design calculations and propulsion analysis across ten propellers |
| Sep 2026 | Scene alert engine with 47 events; Remote Video Terminal; flight replay |
| Sep 2026 | Version 2.0.0 released with a Windows installer |
| Sep 2026 | iDEX Open Challenge application submitted (Ministry of Defence) |
| Oct 2026 | Version 2.0.2: self-updating installs, built-in training simulator |
| Oct 2026 | Aircraft design V1.30, 25-part print split, three-vendor build packages |
| Oct 2026 | Operator manual, first edition |

Nine milestones in four months

PHASE 1 · NOVEMBER 2026 TO APRIL 2027

# Six milestones to a system ready to manufacture and sell

| By end of | Milestone | Proof you can inspect |
|---|---|---|
| Nov 2026 | Orders placed with all three vendors; first airframe parts in hand | Purchase orders, parts on the bench |
| Dec 2026 | Prototype 1 assembled; hover and control checks passed | Hover flight log and video |
| Jan 2027 | First full flight: vertical take-off, cruise, vertical landing | Flight log and video |
| Feb 2027 | Camera payload fitted; prototype 2 built from the same drawings | Two aircraft flying |
| Mar 2027 | Full mission with live AI alerts; endurance and wind tests | 150 min and 13 m/s test results |
| Apr 2027 | Design frozen: production drawings, costed parts list, manuals | Production pack and unit cost |

**End goal of Phase 1: a complete system that is ready for mass manufacturing and for sale.**

Phase 2, from May 2027: first production batch, pilot customers and sales.

THE ASK

# ₹1 crore for 20% equity: a system ready to manufacture and sell

## Use of funds (indicative)

| | |
|---|---|
| Personnel | ₹27 L |
| Airframe and hardware: two prototype aircraft | ₹23 L |
| Testing and flight operations | ₹14 L |
| Camera payload and onboard compute | ₹10 L |
| Lab, workshop and engineering equipment | ₹10 L |
| Company, regulatory, legal and documentation | ₹9 L |
| Ground station and software | ₹7 L |

## What the investor gets

- 20% equity for ₹1 crore
- Funds released in parts, against the six monthly milestones
- By April 2027: two flying aircraft and a frozen, costed design ready for production
- A team ready to start: candidates are already shortlisted

Non-dilutive funding in parallel: iDEX submitted; seed-fund and DRDO schemes next.

Category amounts are indicative and will be refined as vendor quotes come in.

THE FOUNDER

# Built by someone who has flown, fixed and fielded drones for five years



## Aman Singh

**UAV systems engineer · Dehradun**

- Four years at a UAV manufacturer: flight-log analysis, root-cause diagnosis and validation flights across a fleet of 20+ aircraft
- Cut fleet downtime by about 15% with preventive maintenance
- Now leading VTOL development from concept to flight test
- Designed and built a 4 kg mapping quadcopter from scratch, including its circuit board
- Earlier: drone-data and GIS engineering, and an AI crop-health system for a Himalayan conservation NGO

### Let's talk


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Team: candidates for the first hires are already shortlisted.